

The background is a dark, abstract composition featuring various data visualization elements. There are several line graphs with fluctuating lines, bar charts, and a pie chart. A prominent blue geometric shape, resembling a stylized 'L' or a corner of a cube, is positioned on the left side. The overall aesthetic is technical and digital.

SHARC Tank Network Interface Development

Robert Ruiz

Major: Cybersecurity

School: Mercy University

Mentor: Ben Stolz

What is the SHARC LAB?

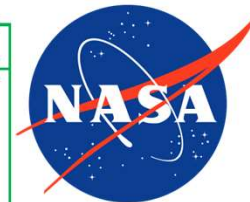
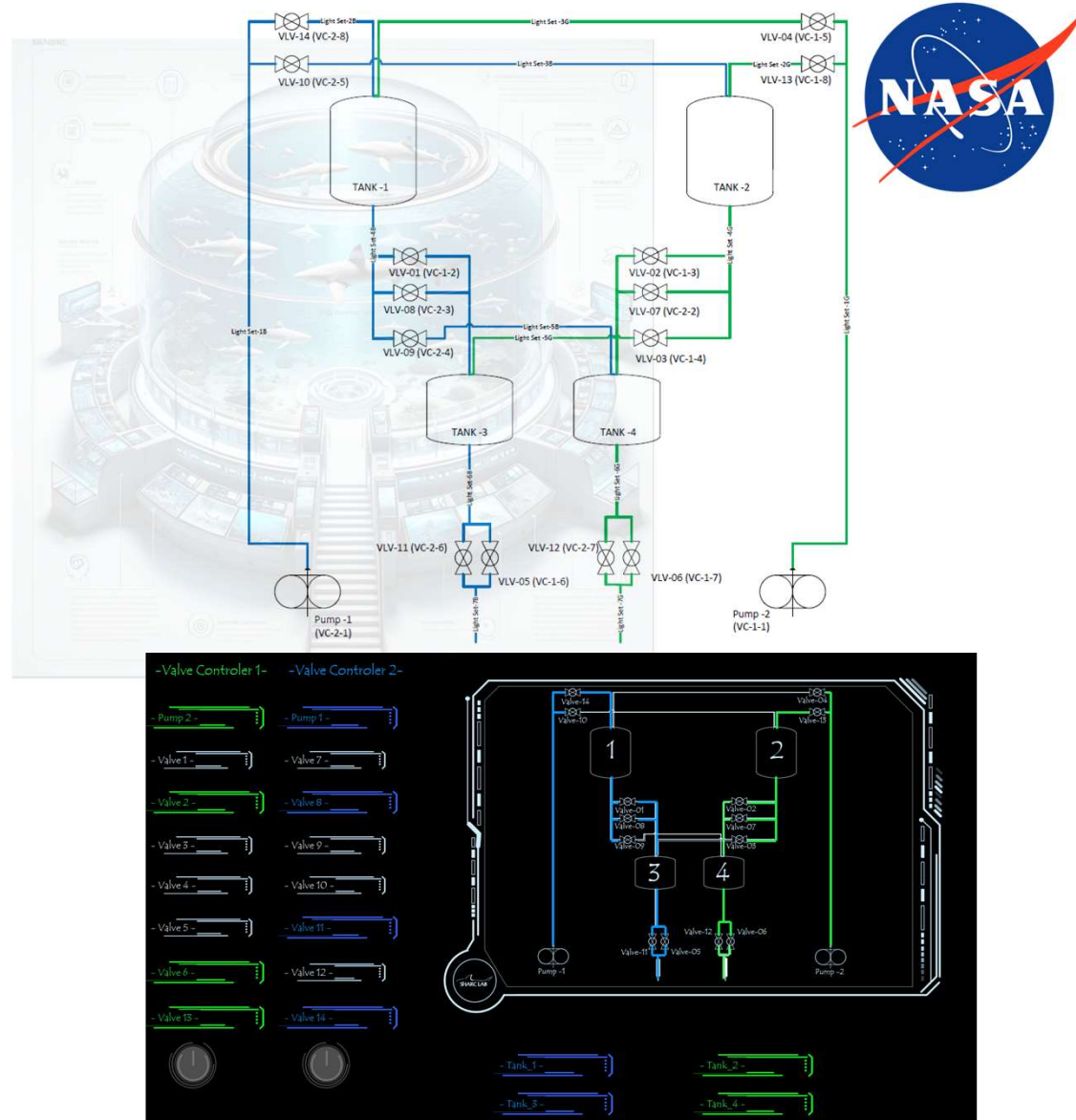
- Systems Hazards Autonomous and Redundancy Controls
- Use to Develop Autonomous Systems





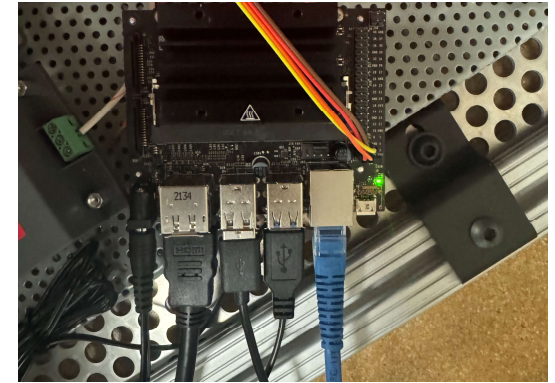
SHARC Tank

- Hardware Simulation of M&FM systems using a GUI
- Non-CUI system that can be shared amongst Different Industries
- Representation of prototypical aerospace system (propulsion, habitat, lunar regolith processing, etc.)



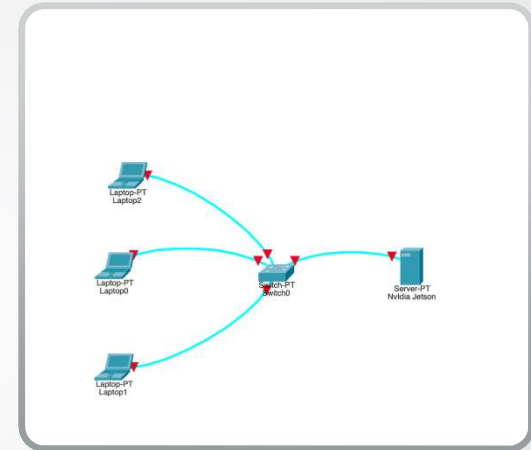
Project Overview

- Setting Up Jetson For Static IP Address
- Interfacing With Sharc Tank



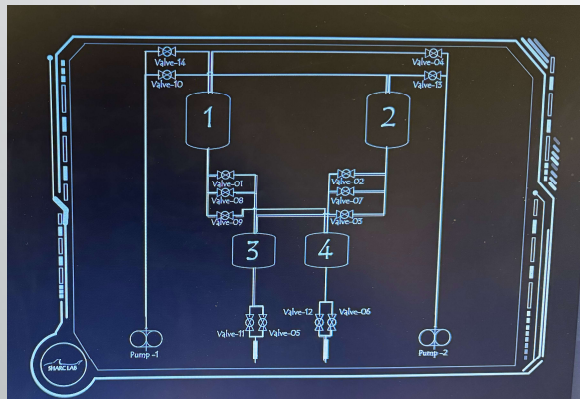
Static IP For Nvidia Jetsons

- Set up Linux Command Lines for interface
- Configured different Static Ips



```
Select ev43projects@ev43projects-desktop: ~  
interface=eth0          # Replace with your network interface  
dhcp-range=169.254.38.80,169.254.38.180,12h # Set the range and lease time  
dhcp-option=option:router,169.254.38.90    # Gateway  
dhcp-option=option:dns,8.8.8.8,8.8.4.4     # DNS server  
  
dhcp-host=00:11:22:33:44:55,169.254.38.100,12h  
dhcp-host=00:11:22:33:44:55,169.254.38.101,12h  
dhcp-host=00:11:22:33:44:55,169.254.38.102,12h  
dhcp-host=00:11:22:33:44:55,169.254.38.103,12h  
dhcp-host=00:11:22:33:44:55,169.254.38.104,12h
```

Demo Of Project



Outcome

- How I helped the branch
 - Develop Lab Network for multiple computer interfaces to the lab hardware (antagonist/protagonist)
 - Created Python Library to interface with SHARC Lab
 - Simplifies future SW Development
 - Develop Ethernet command structure to allow flexibility in SW Dev
 - Facilitates Collaboration with branches across MSFC, Academia, and Small Business Innovation Research
- Skills
 - Networking
 - Linux
 - Python
 - Active Directory



Conclusion

THANK YOU!

